Explainable AI, Fairness and Bias

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With materials from AI BSc course AI for Society

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Announcements

- Final project presentations

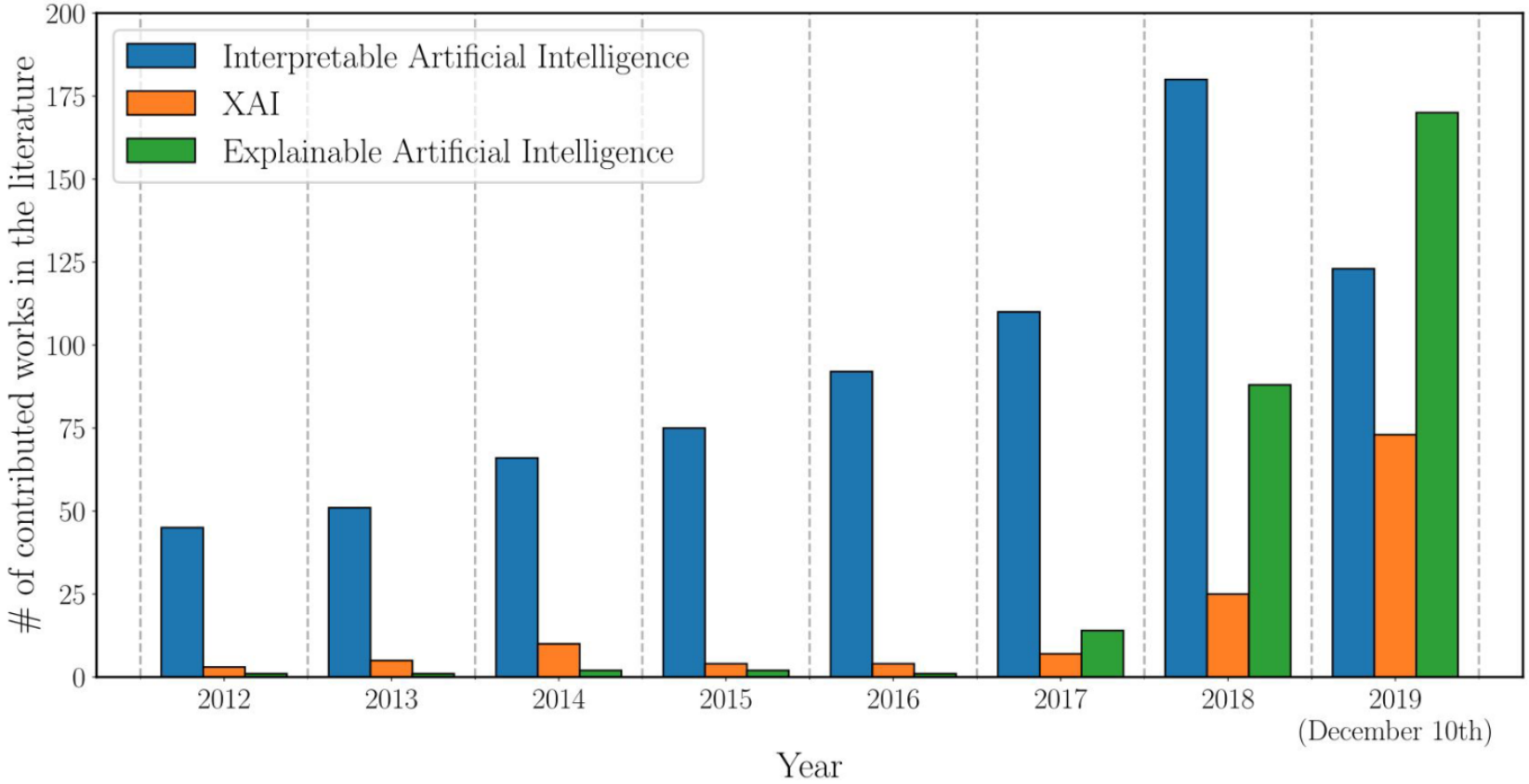
Bias in Word Embeddings: Reading-specific questions

- Why did hard debiasing not completely break the embeddings?
- Bulakbasi et al. argue that in order to perform debiasing, we often have to perform complex computation. In contrast, Bender et al. bring up the ethical risks of using such a computationally heavy model. In general, both issues are related to the ethics of either the use of AI or the implications on society. Is there a way to use a mathematically heavy, but computationally efficient model that satisfies both questions?

Bias in Word Embeddings: Open Questions

- Do modern LLMs (Claude, Gemini) still display the sort of bias we see in the papers?
- Can an LLM like ChatGPT, which is so large ever be fully debiased?
- Should we try to fix biased AI systems by mathematically adjusting them after they are trained, or should we carefully curate their training data from the start so that they never learn the biases in the first place?
- Do you think LLMs should be further developed? Considering what we have learned from the Bender et al. reading. If de-biasing techniques were already known in 2016, and since they seem to work sufficiently well with the trivial examples in the first paper, how come biases are still abundant in large language models such as ChatGPT3?

A recent trend...



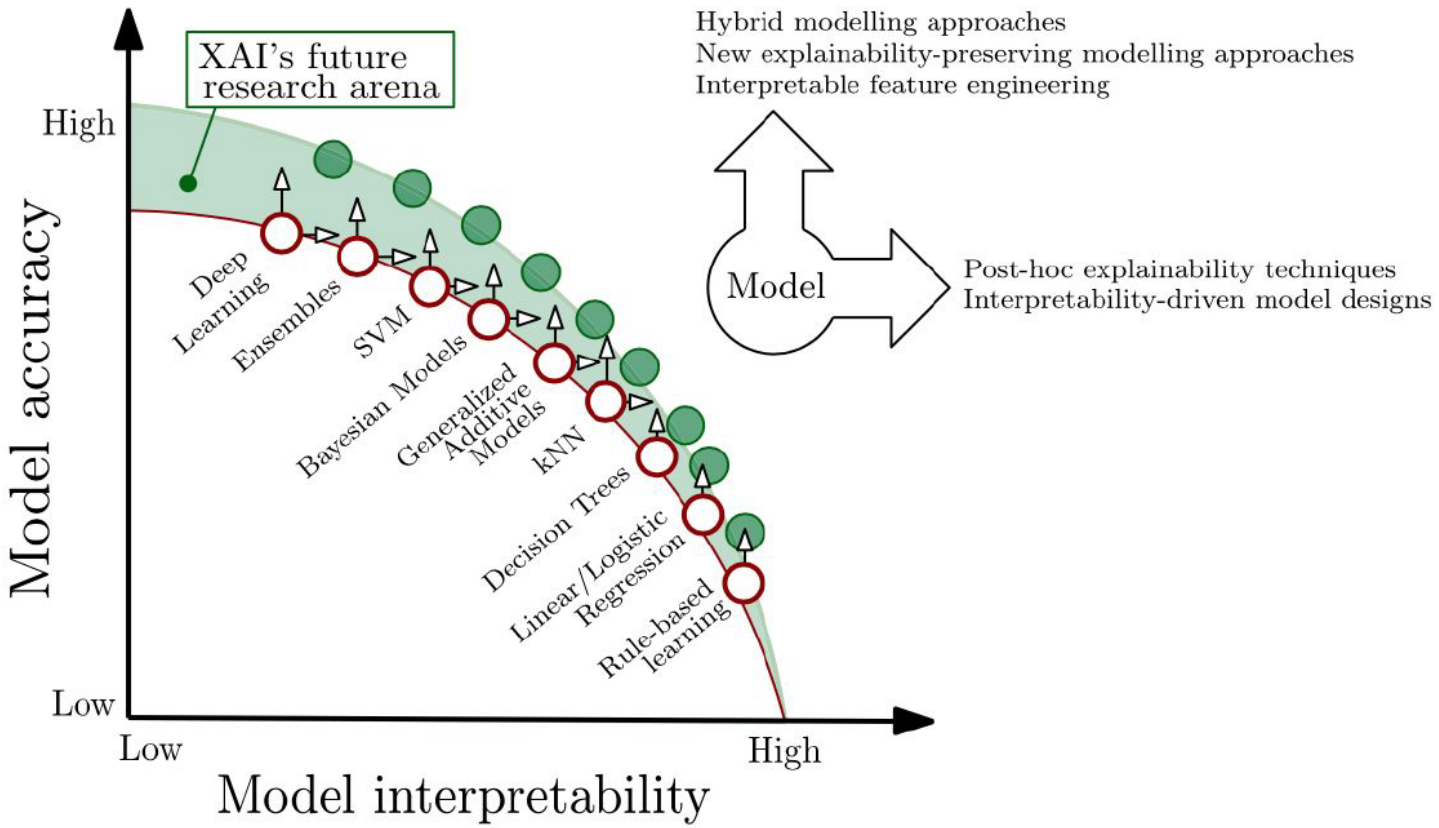
https://arxiv.org/abs/1910.10045

Explainable and Interpretable AI

- The definitions are not very clear yet, as it is an emerging field
- Interpretation: how does a model work? (model transparency)
  - Allow human to grasp the mechanism used to come up with a decision
- Explanation: what can a model tell me? (post-hoc reasoning)
  - Deconstruct steps that were used in making a decision

Explain to whom?

Performance vs Interpretability tradeoff



Social aspects of the explanation/interpretation

- Confidence: grows when the rationale of a decision is close to the thought processes of the user
- Trust: grows when decisions do not require validation to be acted upon
- Safety: the system is consistent and relible, displays uncertainty or confidence level, is robust to outliers etc.
- Ethics: the system does not violate a certain well-defined code of principles

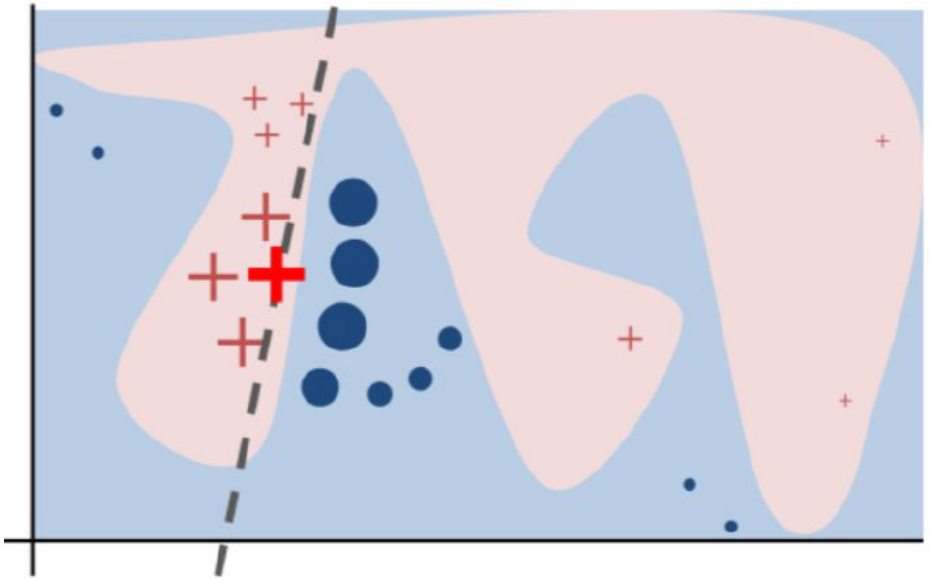
`https://arxiv.org/abs/2004.14545`

Contextual aspects of the explanation/interpretation

- Contrastive: identify elements unique to this decision
- Selective: provide the most relevant causes
- Provide causes: humans are bad at interpreting probabilities
- Social context: may call for different kind of explanation

LIME: Local Interpretable Model-agnostic Explanations

- Algorithm that explains predictions of a classifier by approximating it locally (in the vicinity of the predicted data point) with an interpretable model
- Treat original model as black box
- Train simple interpretable linear classifier on input features and classification decision



https://arxiv.org/abs/1602.04938

Explainable AI

- Can we have explanation without interpretability?
- Can people accurately explain how they make decisions?

Links on Explainable AI

- List of libraries to explain black-box models: `https://github.com/EthicalML/awesome-production-machine-learning#explaining-black-box-models-and-datasets`
- LIME implementation in Python: `https://github.com/marcotcr/lime`
- SHAP unifies LIME and many more methods: `https://github.com/slundberg/shap`
- AIX360: `https://github.com/Trusted-AI/AIX360`
- Language Interpretability Tool (from UvA): `https://github.com/pair-code/lit`

Understanding Language Models: BERTology

- How do you study a black box language model?

https://arxiv.org/pdf/2002.12327.pdf

BERTology questions

- Does BERT base itself on the syntax of human language or just on the linear order of the words?
- Is syntactic structure in the attention weights or in the token representations?
- Does BERT understand negation?
- Does BERT know subject-verb agreement?
- Does BERT understand numbers?
- BERT as a knowledge base?

BERTology methods

- Probing classifiers
  - Use hidden states or attention weights as input to a classifier that predicts a linguistic property of the input text
- Visualization
- Input perturbation
- Masked Language Modeling task
- Nonce word task
- Model perplexity/surprisal

Masked Language Modeling example

AllenNLP Interpret
https://allennlp.org/interpret

Ai2 Allen Institute for AI

AllenNLP

Simple Gradients Visualization

See saliency map interpretations generated by visualizing the gradient.

Saliency Map:

[CLS] The [MASK] rushed to the emergency room to see her patient . [SEP]

Mask 1 Predictions:

47.1% **nurse**

16.4% **woman**

10.0% **doctor**

3.4% **mother**

3.0% **girl**

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Fairness and bias in AI

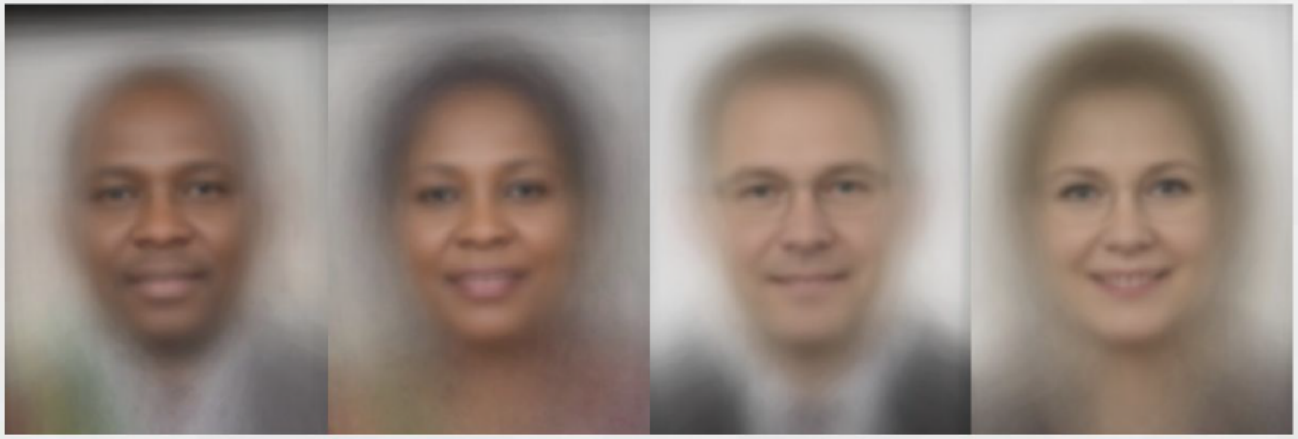
Fairness in AI

- Not a very clearly defined concept
- Lack of bias in decisions
- Balanced treatment of sub-populations and individuals
- Equality of opportunity
- Equity in outcomes

Definition of fairness are often mutually exclusive (mathematically and morally).
Some attempts at formal definitions of fairness in AI:
https://arxiv.org/abs/1901.10002
https://www.annualreviews.org/doi/abs/10.1146/annurev-statistics-042720-125902
https://arxiv.org/abs/1908.09635

Bias in facial recognition

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

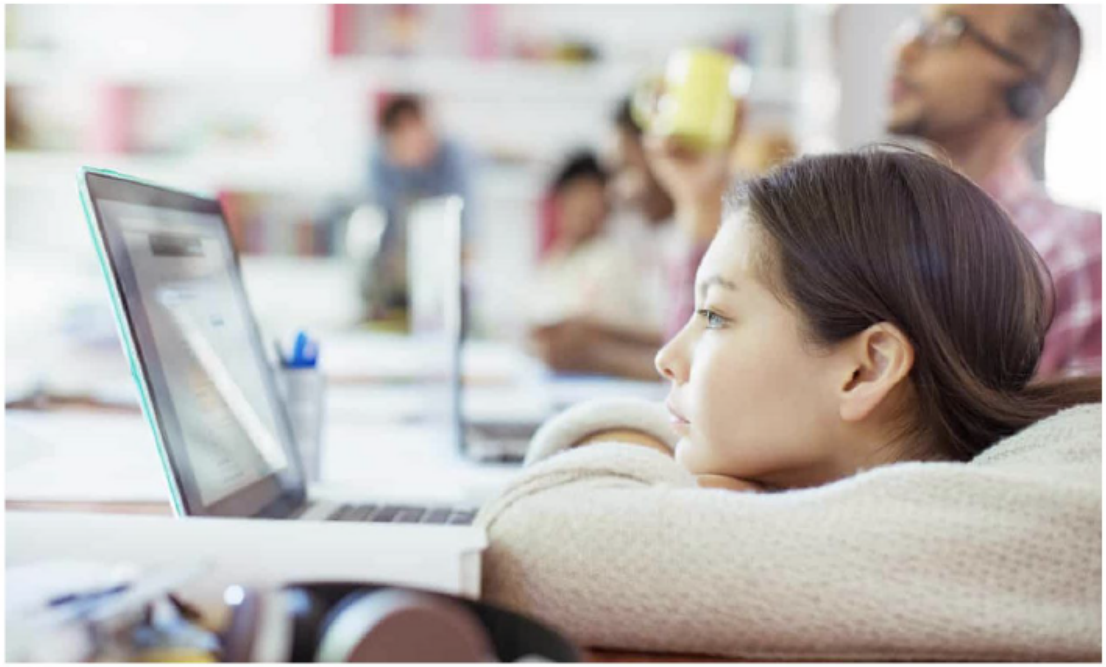


http://gendershades.org/index.html

Bias in job ad recommendation

Women less likely to be shown ads for high-paid jobs on Google, study shows

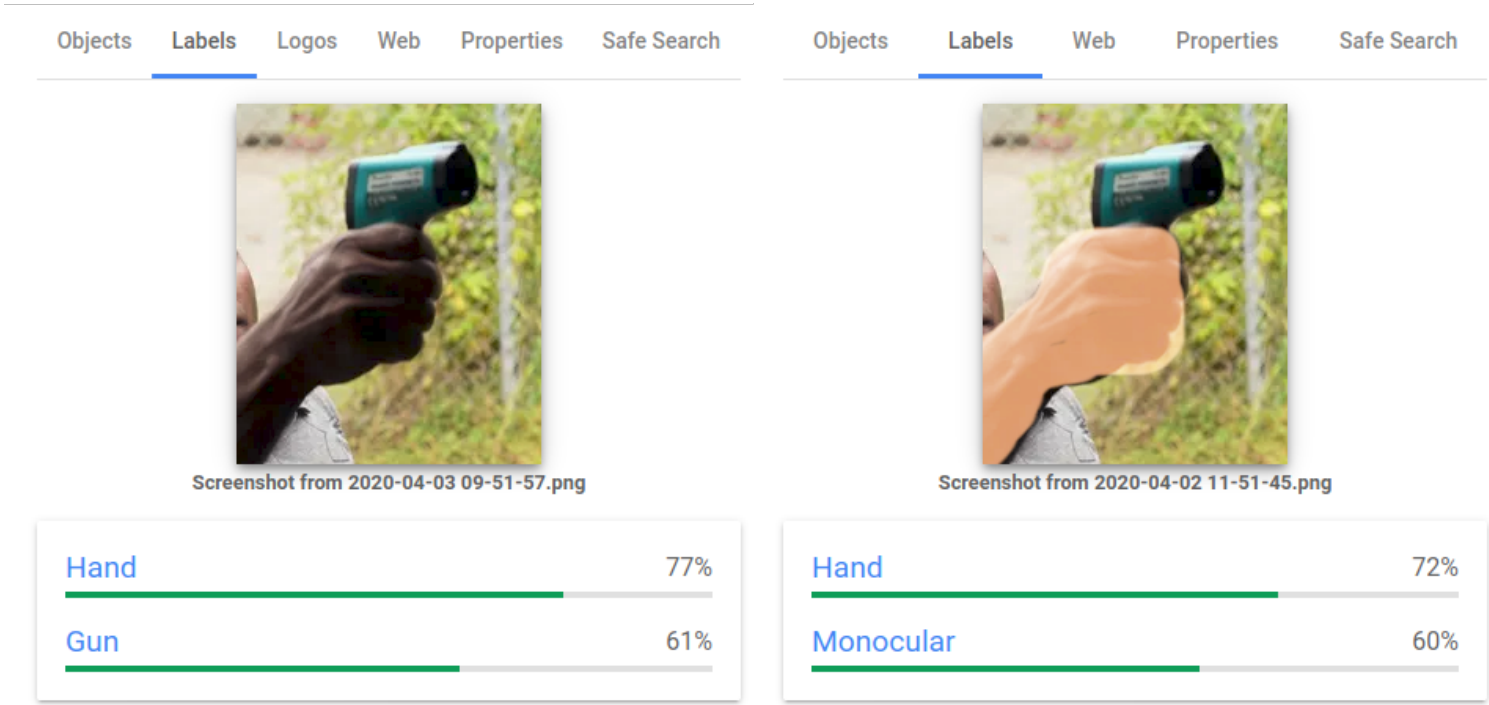
Automated testing and analysis of company's advertising system reveals male job seekers are shown far more adverts for high-paying executive jobs



▲ One experiment showed that Google displayed adverts for a career coaching service for executive jobs 1,852 times to the male group and only 318 times to the female group. Photograph: Alamy

https://www.theguardian.com/technology/2015/jul/08/women-less-likely-ads-high-paid-jobs-google-study

Bias in Google Vision AI

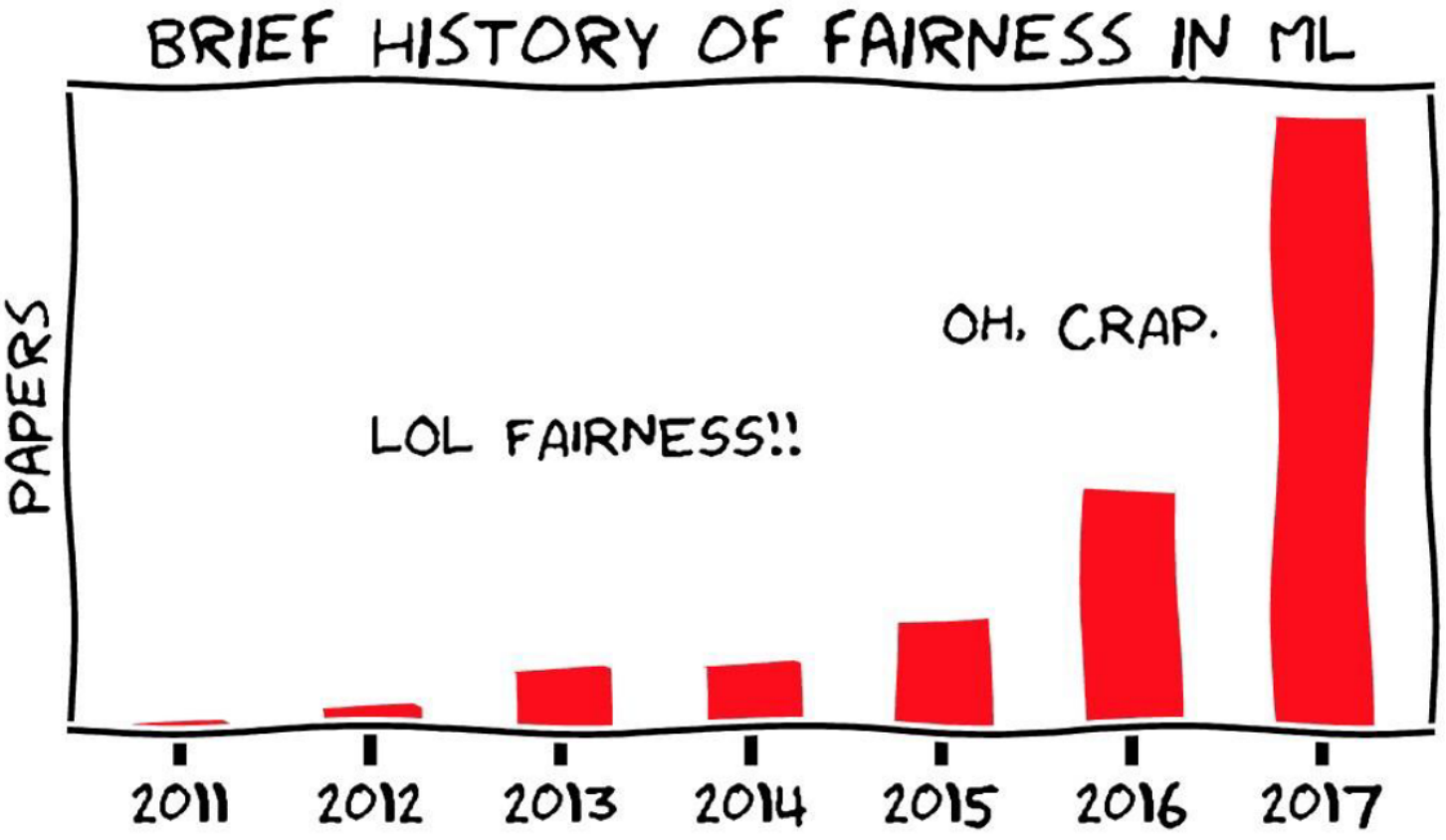


https://algorithmwatch.org/en/google-vision-racism/

Consequences of lack of fairness

- **Impact of error types**: Sometimes a false positive (being falsely recognized as a shoplifter) is worse than a false negative (being falsely flagged as innocent)
- **Disparate impact**: Being flagged as holding a gun by error usually has worse consequences than being flagged holding something else by error.
- **Allocative harm**: Unfair allocation of resources (e.g. hiring decisions)
- **Representational harm**: Unfair depiction of individuals or groups (e.g. stereotyping)

Kate Crawford's lecture 'The trouble with bias':
`https://www.youtube.com/watch?v=fMym_BKWQzk`



Credit: Moritz Hardt

Fairness

- Who is responsible for algorithmic unfairness?

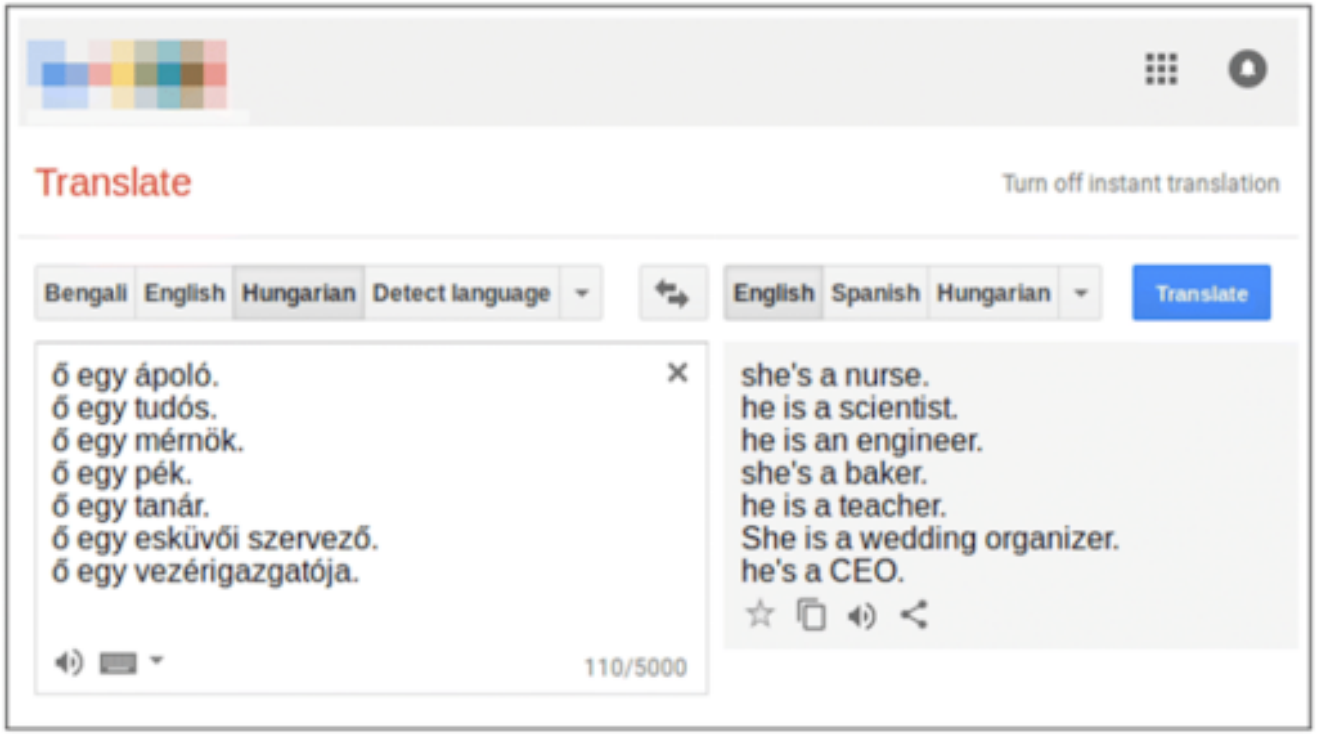
Bias

- Stereotypical bias
- Statistical bias
- Cognitive bias
  - https://upload.wikimedia.org/wikipedia/commons/a/a4/The_Cognitive_Bias_Codex_-_180%2B_biases%2C_designed_by_John_Manoogian_III_%28jm3%29.png

Considering bias in building AI systems

- Define what bias and fairness means in the context of your task
- Explore your data: skewness, outliers, missing values, unbalance across protected groups. Avoid possible bias in data acquisition
- Consider underrepresented and protected groups in model evaluation.
- Consider intersections of protected/underrepresented groups
- Consider possibly unintended consequences when deploying
- Ask for diverse feedback (especially from protected groups involved)

Gender bias and stereotyping



Measuring (gender) bias in word embeddings

- Define a set of "definitional word pairs" that capture the gender dimension (e.g., he/she, man/woman, etc.)
- Measure bias by how differently a word *w* projects onto word pairs.
  - *x_he = cos("politician", "he")*
  - *x_she = cos("politician", "she")*
  - *x_he - x_she* = measure of bias towards the masculine gender

Bolukbasi et al. (2016): `https://arxiv.org/abs/1607.06520`

Measuring (gender) bias in word embeddings

Identify the gender subspace:

- Consider the pairwise differences among the set of "definitional word pairs" that capture the gender dimension (he-she, etc.)
- Apply dimensionality reduction on them (e.g. PCA), and find the gender subspace.
- Use the cosine between any word and this gender subspace to quantify its bias. This bias can be averaged over a set of words.
- If you take masculine - feminine, a positive cosine might be indicative of bias towards the masculine gender, vice versa for a negative one.

Dealing with (gender) bias in word embeddings

- Neutralize and equalize (**hard de-biasing**): enforces that any gender neutral word is set to zero onto the gender subspace.
- Soften (**soft de-biasing**): Ensures that neutral words are equidistant from equality sets. For example, it ensures that brother, sister and husband, wife are both equidistant from babysitting, although probably the latter set will still be closer than the former.

Approaches to bias in word embeddings

1. Work on data (e.g. filtering the training corpus)
2. Work on the algorithm (loss, bias mitigation via a constrained optimization objective)
3. Post-hoc methods (transforming the embeddings in some way)

https://www.aclweb.org/anthology/P19-1159
https://www.aaai.org/AAAI22Papers/AISI-6900.DingL.pdf